Name: ..... Due: **Next Class / Friday**

C3 The Electric Guitar HW 2 – Gibson Les Paul, PAF Humbuckers & Led Zeppelin 'Whole Lotta Love'

Artist: Led Zeppelin (Led Zeppelin II, 1969, Producer/Guitarist: Jimmy Page, Engineer: Eddie Kramer)
https://www.youtube.com/watch?v=HQmmM_qwG4k

1. "Whole Lotta Love" defined heavy blues-rock guitar production. Identify the technical recording components below. (3)

Production Component	Hardware & Technical Implementation
Main Riff Guitar	1959 Gibson Les Paul Standard through 100W Marshall Super Lead head
Middle Psychedelic Breakdown	Theremin, tape echo feedback oscillation, and backwards pre-delay reverb
Sola Sound Tone Bender MKII	
Stereophonic Studio Panning	

2. Explain the physics of Seth Lover's 1957 PAF Humbucker. How does Reverse Wind, Reverse Polarity (RWRP) eliminate 50/60 Hz mains hum while summing the musical string signal? (4)

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3. Why does Gibson use 500 kΩ potentiometers with humbuckers instead of the 250 kΩ potentiometers used on Fender guitars? (3)

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4. Harmonic Nodes & Pickup Placement: Explain why Jimmy Page's Les Paul bridge pickup sounds bright and biting, whereas the neck pickup sounds dark, round, and fluty. (3)

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5. In the middle section of "Whole Lotta Love", Jimmy Page used backwards echo. Explain how an engineer creates backwards tape echo in an analog tape recording studio. (3)

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6. Semi-Hollow (ES-335) vs Solidbody (Les Paul) Acoustic & Electrical Properties: compare below: (4)

Gibson ES-335 Centre-Block Architecture & Acoustic Response (2)
Solid maple centre block, hollow acoustic wings, woody acoustic resonance with feedback suppression:
Gibson Les Paul Solid Mahogany/Maple Architecture & Sustain (2)
High-density solid construction, maximum acoustic mass, extended note sustain, and high-gain stability:

Total:	**/18**